Acquirer - all six functions:

1. **Merchant underwriting** — evaluating merchants before onboarding.
2. **Transaction routing** — sending auth requests to the card network and back.
3. **Settlement** — moving actual money to the merchant's account.
4. **Chargeback management** — handling disputes and representment
5. **Compliance and monitoring** — keeping the portfolio within card network rules.
6. **Interchange and fee management** — collecting MDR, passing interchange to issuer.

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1. Merchant Underwriting

What it is

Before a merchant can process a single card transaction, the acquirer must decide: *should I let this business onto my platform?* This is underwriting — a risk and credit decision made before any business relationship begins.

The acquirer is taking on financial liability for every merchant they onboard. If a merchant processes fraudulent transactions, disappears overnight, or can't cover chargebacks, the acquirer absorbs the loss. Visa and Mastercard don't care about the merchant — they come to the acquirer for resolution.

What they actually evaluate

Business legitimacy:

- Is this a real business with a verifiable address, registration, and ownership?
- Who are the beneficial owners? Do any of them appear on OFAC sanctions lists or the MATCH list (terminated merchant file)?
- Does the business model make sense — does the website match the MCC they're applying under?

Financial risk profile:

- What is the expected monthly processing volume?
- What is the average transaction size?
- What is the chargeback risk of this business category? A firearms dealer has a very different risk profile than a grocery store
- Does the business have a history of excessive chargebacks with a prior acquirer?

Fraud risk profile:

- Is the business in a high-risk MCC (merchant category code)? Travel, adult content, nutraceuticals, cryptocurrency — all flagged as elevated risk

- Is the business model inherently prone to friendly fraud? Subscription businesses, for example, generate more chargebacks than one-time purchases
- Does the business have a delivery delay between payment and fulfillment? A merchant selling custom furniture paid upfront but delivered in 12 weeks has high dispute exposure

Velocity and volume risk:

- Is the requested processing volume plausible for a business this size?
- A brand new business requesting $5 million per month processing limit is a red flag
- Volume much higher than industry average for the business type triggers scrutiny

The decision output

The underwriter produces:

- Approve / conditionally approve / decline
- Processing limits (monthly volume cap, per-transaction cap)
- Reserve requirements — for high-risk merchants, the acquirer holds back a percentage of each payout as a rolling reserve against future chargebacks
- Monitoring triggers — thresholds that will automatically flag the account for review

How this maps to your current role

This is the closest analog to what you do now as a Risk Analytics Manager in merchant services. The difference at Stripe is scale — they onboard thousands of merchants per day, making manual underwriting impossible. ML models make the approve/decline decision in real time at signup, with human review reserved only for edge cases the model flags as uncertain.

The ML version at Stripe

- Binary classification: approve vs. decline at onboarding
- Regression: predicted chargeback rate, predicted fraud rate
- Features: business type, MCC, geographic location, website content analysis, owner identity signals, requested volume vs. industry benchmarks, device fingerprint of the signup session
- Ground truth labels: actual chargeback rates, fraud rates, and early terminations observed 90–180 days after onboarding

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2. Transaction Routing

What it is

Every card transaction needs to travel from the merchant's terminal or website through the payment infrastructure to the issuing bank and back in under one second. The acquirer is responsible for making that journey happen correctly and efficiently.

The technical flow in detail

```
Customer taps card at merchant terminal
        ↓
Terminal/gateway formats authorization request message (ISO 8583 standard)
        ↓
Message sent to acquirer's processing system
        ↓
Acquirer determines which network to route to (Visa? Mastercard? Debit
network?)
        ↓
Acquirer formats and submits to card network
        ↓
Card network identifies issuer via BIN, routes to issuer
        ↓
Issuer's authorization system evaluates: approve or decline?
        ↓
Response travels back: Issuer → Network → Acquirer → Merchant
        ↓
Merchant receives approve/decline in ~500ms
```

What the acquirer decides during routing

Network selection — many cards can run on multiple networks. A Visa debit card can often be routed through Visa or through a regional debit network like Star, Pulse, or NYCE. The rates differ. The authorization approval rates differ. The acquirer chooses.

Data enrichment — the acquirer can add data fields to the authorization request that improve approval odds. Sending AVS (address verification), CVV, and 3DS authentication data tells the issuer this transaction has been verified, reducing their fraud risk perception and increasing approval probability.

Transaction presentation — how a transaction is formatted and categorized affects both the interchange rate it qualifies for and how the issuer's fraud model scores it. A transaction presented as card-present qualifies for lower interchange than card-not-present, even for the same merchant.

Retry logic — if a transaction is declined, the acquirer can retry it. But not all declines are equal. A "do not honor" decline is different from an "insufficient funds" decline, which is different from a "lost/stolen card" decline. Smart retry logic reads the decline reason code and decides whether to retry immediately, retry after a delay, try a different network, or give up.

The ML version at Stripe

This is where some of Stripe's most valuable ML lives:

Authorization rate optimization model — predicts the probability of approval for each transaction on each possible network/presentment combination. Routes to maximize approval probability for legitimate transactions.

Decline classification model — when a transaction is declined, classifies the decline reason and predicts whether a retry will succeed, on what network, and when.

Network selection model — for dual-network cards, predicts which network will yield better approval rate at lower interchange cost.

Real-time fraud scoring (Radar) — runs in parallel with routing, scoring every transaction for fraud probability. A high fraud score triggers additional friction (3DS challenge) or decline before the transaction even reaches the issuer.

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3. Settlement

What it is

Authorization and settlement are two completely separate processes that most people conflate. Authorization is a promise — the issuer saying "yes, funds are available and reserved." Settlement is the actual movement of money.

The settlement timeline

```
Day 0:  Customer makes purchase → Authorization approved
        Money is reserved on customer's account but not moved

Day 1:  Merchant closes their batch (end of business day)
        All authorized transactions submitted for settlement

Day 1-2: Clearing process
        Acquirer submits batch to card network
        Card network matches authorizations with settlements
        Card network calculates net positions between all acquirers and
issuers

Day 2:  Settlement
        Issuer transfers funds to card network
        Card network transfers to acquirer
        Acquirer deposits to merchant's bank account minus fees
```

What actually moves in settlement

Not individual transaction amounts — the system is far more efficient than that. The card network calculates **net settlement positions**:

Chase (as issuer) owes Stripe (as acquirer) $10 million today across all Stripe merchants whose customers used Chase cards. Stripe (as acquirer) owes Chase $3 million for Stripe cardholders who used Chase-acquired merchants. Net: Chase wires Stripe $7 million. One wire, billions of transactions settled.

Payout to merchants

After the acquirer receives settlement funds, they distribute to merchants minus:

- Interchange (passed through to issuers)
- Assessment fees (passed through to card networks)
- Acquirer processing margin (kept)
- Any reserve holdbacks for high-risk merchants

Stripe's standard payout is T+2 (2 business days after transaction). They offer instant payouts for an additional fee — which requires Stripe to advance the funds before receiving settlement, taking on the float risk themselves.

The risk in settlement: bust-out fraud

The most dangerous settlement risk is a merchant who processes a large volume of legitimate-looking transactions, receives their payouts from the acquirer, then disappears before the chargebacks arrive 60–90 days later. The acquirer has paid out funds they'll never recover.

This is called bust-out fraud and it's a major reason why:

- High-risk merchants have rolling reserves
- Transaction volume spikes trigger review
- Payout timing is sometimes delayed for new merchants

The ML version at Stripe

Payout fraud model — predicts the probability that a merchant will bust out before chargebacks arrive. Features include: payout acceleration requests, sudden volume spikes, product category changes, customer dispute rate trends, time since onboarding.

Float optimization model — for instant payouts, Stripe is lending money. ML predicts which merchants are safe to advance funds to and which carry too much settlement risk to offer the feature.

Reserve calibration model — how much reserve to hold for each merchant, dynamically adjusted as their risk profile changes over time.

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4. Chargeback Management

What it is

A chargeback is a forced transaction reversal initiated by the cardholder's bank. The acquirer sits in the middle of every chargeback, managing the flow between the issuer who initiated it and the merchant who must respond.

The full chargeback lifecycle

```
Step 1 — Cardholder disputes with issuer
Cardholder calls Chase: "I didn't make this charge"
Chase investigates briefly, provisionally credits cardholder

Step 2 — Issuer files chargeback
Chase files chargeback through Visa network
Chargeback reason code assigned (e.g., 10.4 = card absent fraud)

Step 3 — Acquirer receives chargeback
Stripe receives notification
Funds debited from merchant's account or reserve

Step 4 — Acquirer notifies merchant
Stripe informs merchant: you have 7-30 days to respond
Merchant decides: accept the chargeback or fight it

Step 5 — Representment (if merchant fights)
Merchant submits evidence to Stripe
Stripe formats and submits representment to Visa
Visa forwards to Chase

Step 6 — Issuer reviews representment
Chase reviews merchant's evidence
Decision: uphold chargeback or reverse in merchant's favor

Step 7 — Pre-arbitration (if issuer upholds)
Merchant can escalate to pre-arbitration
Both sides submit final arguments

Step 8 — Arbitration (last resort)
Visa/Mastercard makes final binding decision
Losing party pays arbitration fees ($500+)
```

Chargeback reason codes

Every chargeback has a reason code that tells you why it was filed. These codes are defined by the card networks and are critical ML features:

Fraud codes (most common and serious):

- 10.4 — Card absent environment (online fraud, card not physically present)
- 10.1 — EMV liability shift (chip card used on non-chip terminal)

Authorization codes:

- 11.1 — Card recovery bulletin (card was on a block list)
- 11.3 — No authorization (merchant didn't get proper auth)

Consumer dispute codes:

- 13.1 — Merchandise not received
- 13.2 — Cancelled recurring transaction
- 13.3 — Not as described
- 13.7 — Cancelled merchandise/services (friendly fraud hotspot)

Chargeback thresholds — why this is existential for acquirers

Visa and Mastercard set strict thresholds for chargeback rates. Exceed them and consequences escalate:

```
Visa thresholds (per merchant):
Standard:   > 0.9% chargeback ratio OR > 100 chargebacks/month
           → Merchant placed in monitoring program, acquirer fined

Excessive:  > 1.8% chargeback ratio OR > 1000 chargebacks/month
           → Higher fines, potential termination

If acquirer's overall portfolio exceeds thresholds:
           → Acquirer fined, potentially losing network membership
```

This is why acquirers terminate merchants proactively — one large merchant with a 3% chargeback rate can endanger the acquirer's entire network relationship.

The ML version at Stripe

Chargeback prediction model — predicts which transactions will result in chargebacks before they happen. Used to decline high-risk transactions in real time (Radar) and to flag merchants approaching threshold violations.

Representment outcome model — predicts whether fighting a specific chargeback will succeed given the reason code, evidence available, and issuer behavior patterns. Helps merchants decide which chargebacks are worth fighting.

Friendly fraud detection — distinguishes chargebacks that represent genuine fraud from those where the cardholder received goods/services but filed a dispute anyway. Friendly fraud chargebacks have different features: the cardholder has a history of disputes, the merchant has delivery confirmation, the dispute was filed very close to the chargeback time limit.

Threshold monitoring model — tracks each merchant's rolling chargeback rate in real time and predicts trajectory. Triggers proactive intervention before the merchant hits Visa/Mastercard thresholds.

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5. Compliance and Monitoring

What it is

Onboarding a merchant is not a one-time event. The acquirer is continuously responsible for everything that merchant does on their platform. Compliance monitoring is the ongoing surveillance of the entire merchant portfolio for violations, risk changes, and prohibited activity.

What the acquirer monitors

Chargeback rates — as covered above, the most visible compliance metric. Tracked daily per merchant and in aggregate across the portfolio.

Transaction laundering — one of the most serious violations. A merchant processes transactions on behalf of another undisclosed business. Example: a legitimate restaurant processes card payments for an unlicensed gambling operation run by the same owner. The card network sees a restaurant; the real activity is hidden. Also called factoring.

Money laundering — merchants used to clean illicit funds. Structured transactions (many small charges just below reporting thresholds), sudden volume spikes with no business explanation, transactions with no plausible economic purpose.

Prohibited business activity — merchants who shift into prohibited categories after onboarding. A supplements merchant who starts selling unapproved pharmaceuticals. A marketplace that starts facilitating illegal goods.

MCC miscoding — merchants deliberately miscoding their business category to get lower interchange rates or to obscure the nature of their business.

Velocity anomalies — sudden changes in transaction volume, average ticket size, refund rate, or geographic distribution that suggest something has changed in the business.

PCI-DSS compliance — all merchants must maintain Payment Card Industry Data Security Standards. The acquirer is responsible for ensuring their merchants are compliant and for reporting violations.

OFAC and sanctions screening — ongoing screening of merchant owners and transaction counterparties against sanctions lists. Not just at onboarding — lists change and new individuals get added.

The card network's role in monitoring

Visa and Mastercard run their own monitoring programs on top of the acquirer's:

- **Visa Fraud Monitoring Program (VFMP)** — flags merchants with excessive fraud rates

- **Visa Dispute Monitoring Program (VDMP)** — flags merchants with excessive dispute rates
- **Mastercard Excessive Chargeback Program (ECP)** — similar thresholds, similar consequences

When a merchant appears in these programs, fines cascade through: network fines the acquirer, acquirer passes fines to merchant. Escalating monthly fines that can reach $100,000+ per month for chronic violators.

The ML version at Stripe

Ongoing merchant risk scoring — every merchant gets a risk score updated continuously as new transaction data arrives. Score changes trigger automated actions: increase reserves, restrict payout frequency, require additional verification, or flag for human review.

Transaction laundering detection — looks for structural patterns that indicate a merchant is processing for undisclosed businesses. Features: unusual product diversity, transaction patterns inconsistent with stated MCC, multiple card-present transactions at implausible geographic locations.

Behavioral anomaly detection — flags when a merchant's behavior deviates significantly from their own historical baseline or from similar merchants in their cohort. This is an unsupervised ML problem — you're detecting what's different, not what matches a known fraud pattern.

Network pattern analysis — because Stripe sees across all merchants, they can detect coordinated activity. A fraud ring running transactions across 50 different sub-merchants simultaneously is invisible at the individual merchant level but obvious in aggregate.

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6. Interchange and Fee Management

What it is

The acquirer sits at the center of a complex fee flow. They collect fees from merchants and distribute them upstream to card networks and issuers. Managing this flow accurately and optimally is both an operational and a strategic function.

The fee collection mechanics

The acquirer collects the Merchant Discount Rate (MDR) from each merchant — typically bundled as a single percentage plus a per-transaction fee. Behind that MDR are three components:

```
MDR collected from merchant:          2.9% + $0.30
```

```
Distributed as:
→ Interchange to issuer:             ~1.80% + $0.10
→ Assessment to card network:         ~0.13%
→ Acquirer margin (Stripe keeps):     ~0.97% + $0.20
```

Interchange qualification

Not all transactions qualify for the same interchange rate — as we covered earlier, there are hundreds of rate categories. The acquirer is responsible for ensuring transactions are submitted with the correct data to qualify for the lowest applicable interchange rate, which directly affects their margin.

A transaction submitted without address verification data might qualify for a higher "non-qualified" interchange rate instead of the standard rate. The difference can be 50–100 basis points. At Stripe's scale, that is hundreds of millions of dollars annually.

Pricing models

Interchange-plus pricing — the merchant pays actual interchange cost plus a fixed acquirer markup. Transparent, fair, better for high-volume merchants.

Flat rate pricing — what Stripe offers retail customers (2.9% + $0.30). Simple, predictable, but the acquirer takes the interchange risk — if a merchant's customers use a lot of premium rewards cards, the actual interchange cost can eat into or exceed the acquirer's margin.

Tiered pricing — buckets transactions into qualified, mid-qualified, and non-qualified tiers. Common with traditional acquirers, criticized for opacity.

The float and timing economics

The acquirer collects interchange from merchants at the point of settlement but doesn't pay it to issuers until the card network's settlement cycle completes. This creates a timing float — the acquirer holds funds for 24–48 hours. At Stripe's transaction volume, even one day of float on interchange flows is a meaningful amount of money.

The ML version at Stripe

Interchange optimization model — for each transaction, determines the optimal combination of network routing, data fields, and presentment format to qualify for the lowest interchange tier while maintaining high authorization rates.

Pricing optimization model — for enterprise merchants, Stripe offers custom pricing. ML models predict the true cost-to-serve each merchant (based on their card mix, chargeback rate, dispute rate) to set pricing that is competitive while maintaining margin.

Revenue forecasting model — predicts total interchange costs, assessment fees, and net revenue across the portfolio. Critical for financial planning at Stripe's scale of $1.4 trillion annual payment volume.

Margin anomaly detection — flags when a merchant's actual interchange cost diverges significantly from predicted cost, indicating a card mix shift, unusual transaction patterns, or potential miscoding.

Putting It All Together

The six acquirer functions are not independent — they form a continuous cycle:

```
Underwriting → sets the risk parameters for who enters the portfolio
        ↓
Transaction Routing → optimizes every transaction that flows through
        ↓
Settlement → moves money and creates the financial record
        ↓
Chargeback Management → resolves disputes and refines fraud labels
        ↓
Compliance Monitoring → continuously re-evaluates every merchant
        ↓
Fee Management → optimizes the economics of the entire operation
        ↓
feeds back into → Underwriting (portfolio learnings improve future decisions)
```

At a traditional acquirer, each of these functions is a separate team, largely manual, operating on weekly or monthly cycles. At Stripe, all six run simultaneously, in real time, powered by ML, across millions of merchants. That is the fundamental difference — and it is why the Payments ML Accelerator team exists.

Your background gives you intuitive understanding of all six functions from the merchant services side. The two-year plan converts that intuition into the ML fluency to build the systems that automate them.